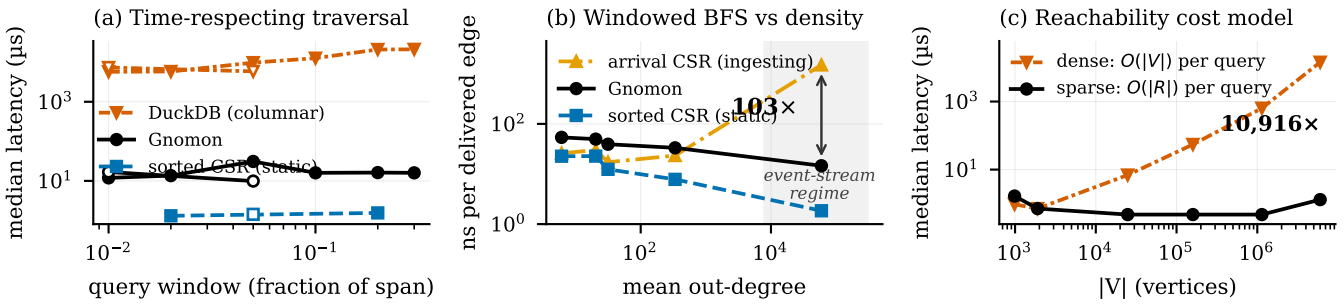
(a) Time-respecting traversal
median latency (µs)
10^3
10^1
DuckDB (columnar)
Gnomon
sorted CSR (static)
10^{-2}
10^{-1}
query window (fraction of span)
(b) Windowed BFS vs density
ns per delivered edge
10^2
10^0
arrival CSR (ingesting)
Gnomon
sorted CSR (static)
103×
event-stream regime
10^2
10^4
mean out-degree
(c) Reachability cost model
median latency (µs)
10^3
10^1
dense: $O(|V|)$ per query
sparse: $O(|R|)$ per query
10,916×
10^3
10^4
10^5
10^6
|V| (vertices)